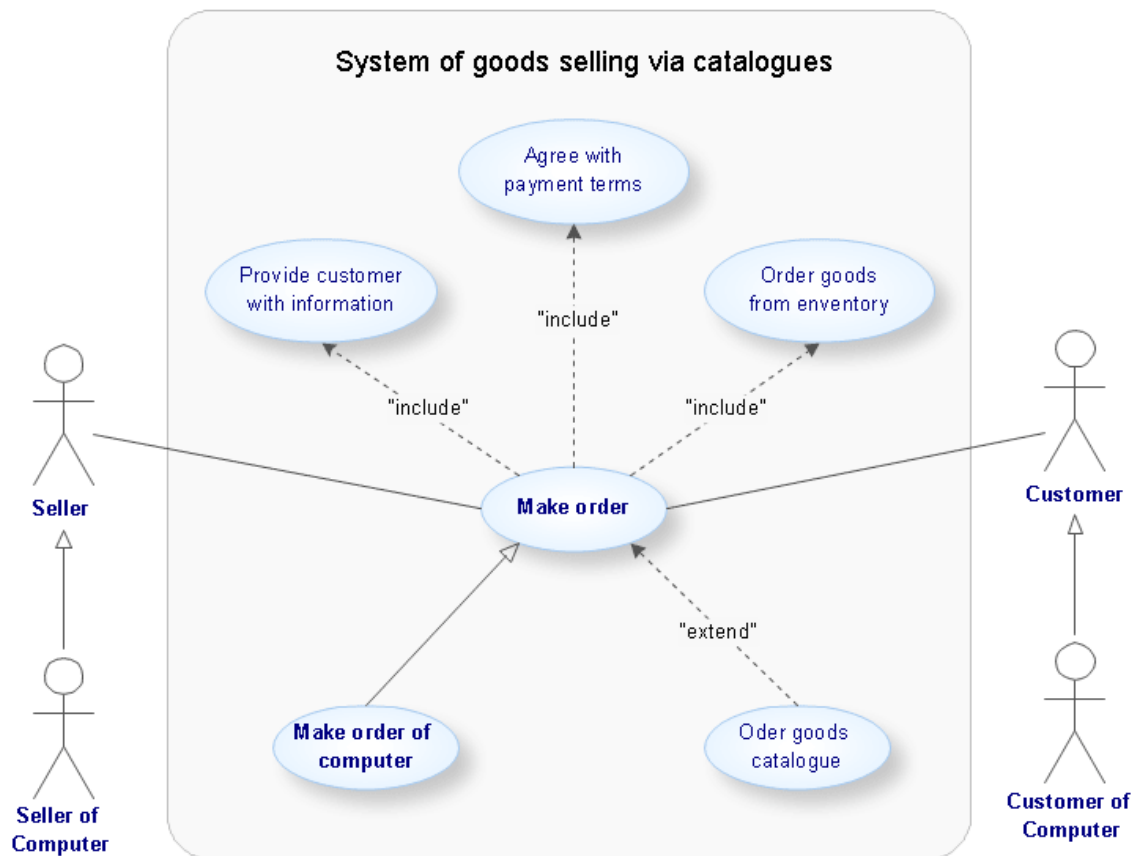


The project will contain a Word document with the following components:

1. **Project title and description**
2. **UML Use Case diagram(s)**

Example



All the use cases will be described with the actor action

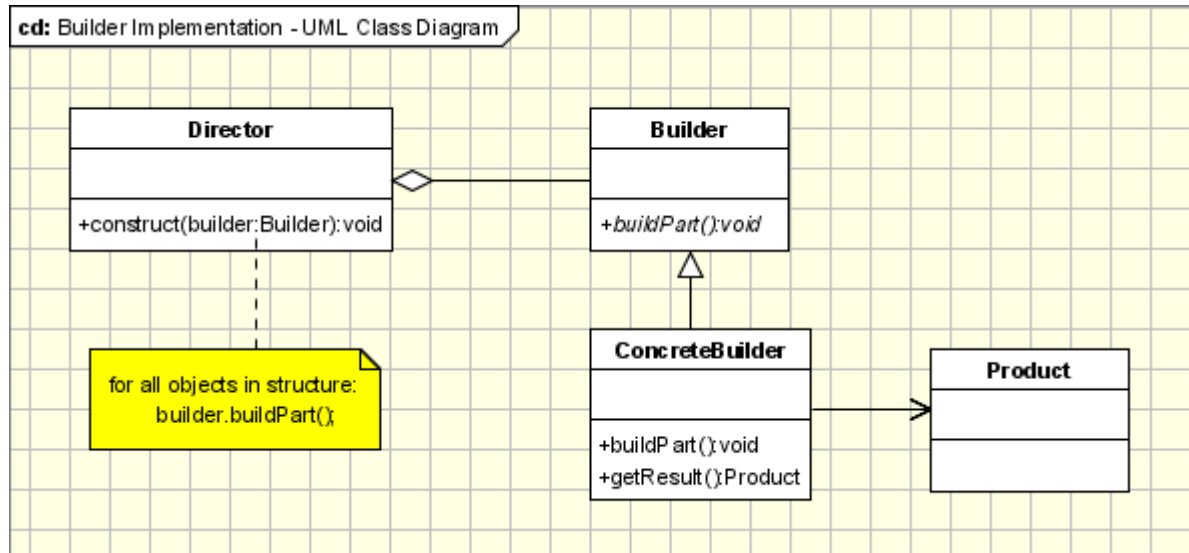
3. **Design pattern and the motivation of using it**

Description of design patterns can be found at the following link: <http://www.oodeesign.com/>

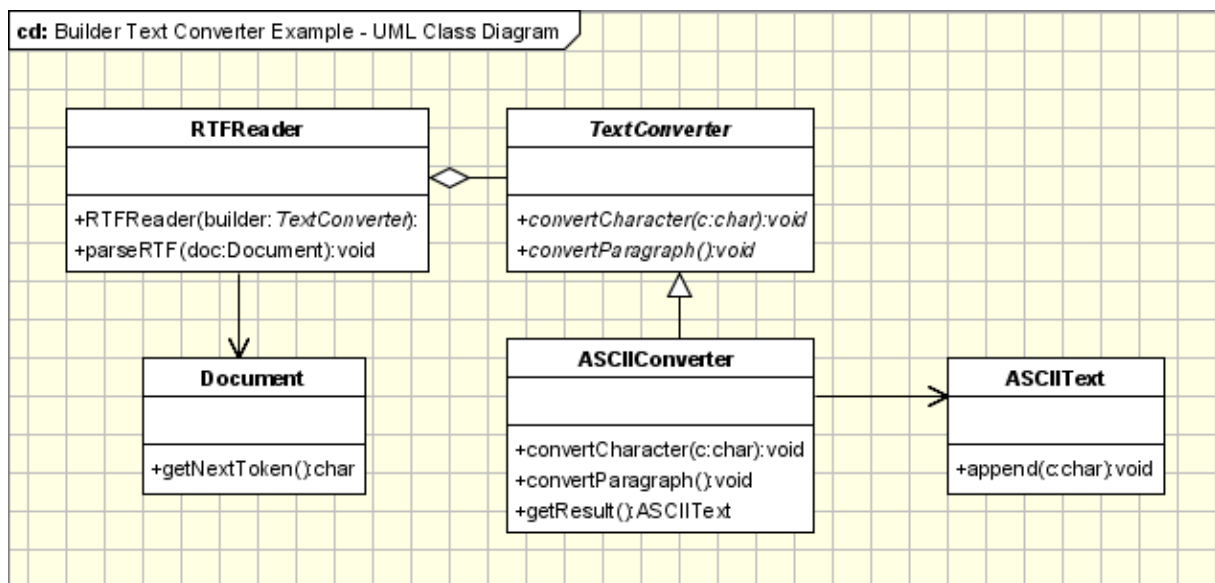
4. UML Class diagrams

After choosing the design pattern , this will be used to create the UML Class diagram
Example

Class diagram for Builder

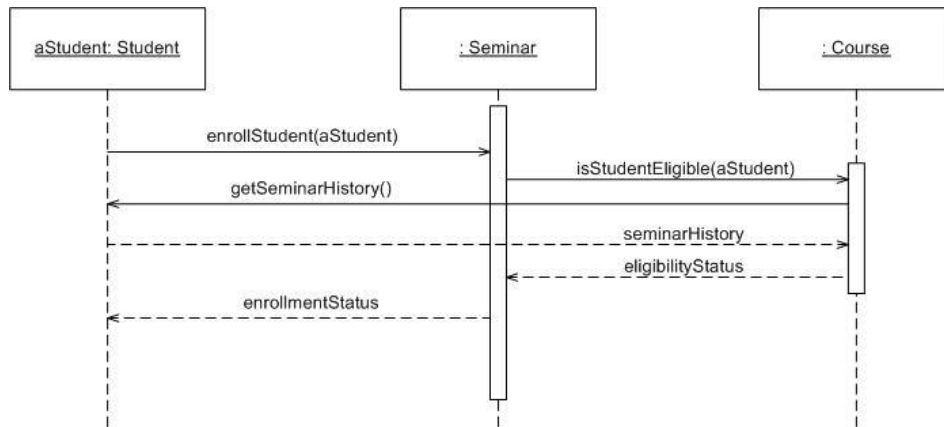


Example for using this design pattern – application for converting a text



5. UML Sequence diagrams (for main communication actions)

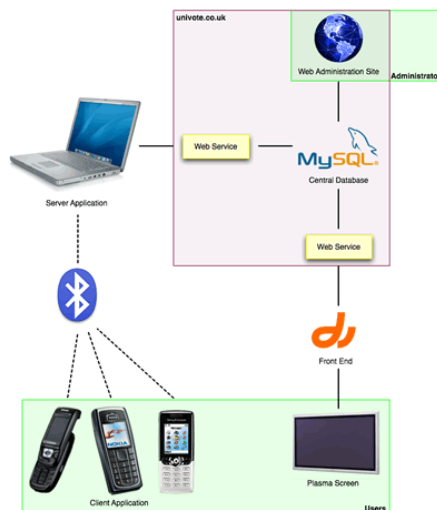
Example



In the document will be describe the actions from Sequence Diagram using a table

Interactions	Description
Interaction 1	
Interaction 2	
Interaction 3	
Interaction 4	

6. Application architecture (showing the modules of the application) and describing this modules



7. The generated code from class diagram (you can choose the language)

Example of generated code - application for converting a text

```
//Abstract Builder
class abstract class TextConverter{
    abstract void convertCharacter(char c);
    abstract void convertParagraph();
}

// Product
class ASCIIText{
    public void append(char c){ //Implement the code here }
}

//Concrete Builder
class ASCIIConverter extends TextConverter{
    ASCIIText asciiTextObj;//resulting product

    /*converts a character to target representation and appends to the resulting*/
    object void convertCharacter(char c){
        char asciiChar = new Character(c).charValue();
        //gets the ascii character
        asciiTextObj.append(asciiChar);
    }
    void convertParagraph(){ }
    ASCIIText getResult(){
        return asciiTextObj;
    }
}

//This class abstracts the document object
class Document{
    static int value;
    char token;
    public char getNextToken(){
        //Get the next token
        return token;
    }
}

//Director
class RTFReader{
    private static final char EOF='0'; //Delimiter for End of File
    final char CHAR='c';
    final char PARA='p';
    char t;
    TextConverter builder;
    RTFReader(TextConverter obj){
        builder=obj;
    }
    void parseRTF(Document doc){
        while ((t=doc.getNextToken())!= EOF){
            switch (t){
                case CHAR: builder.convertCharacter(t);
                case PARA: builder.convertParagraph();
            }
        }
    }
}

//Client
public class Client{
    void createASCIIText(Document doc){
        ASCIIConverter asciiBuilder = new ASCIIConverter();
        RTFReader rtfReader = new RTFReader(asciiBuilder);
        rtfReader.parseRTF(doc);
        ASCIIText asciiText = asciiBuilder.getResult();
    }
}
```

```
    }  
    public static void main(String args[]){  
        Client client=new Client();  
        Document doc=new Document();  
        client.createASCIIText(doc);  
        system.out.println("This is an example of Builder Pattern");  
    }  
}
```

- 8. Print screens from the developed project or pictures**
- 9. All the codes, diagrams, resources used in the project and the Word document will be uploaded in the Repository.**